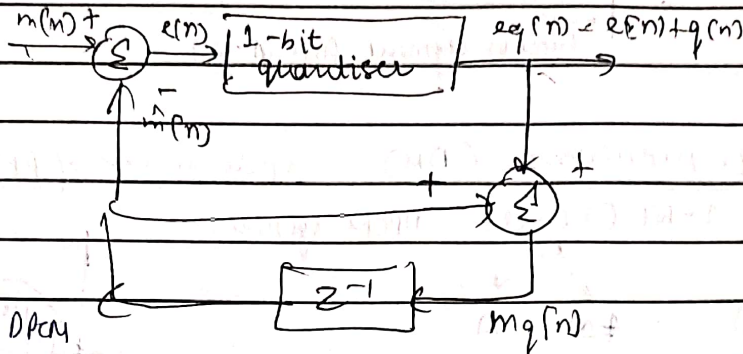


$\Delta = \text{range} / 2$

$$P_s = \text{avg. signal power} = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$\sigma_q^2 = A^2/12$$

Delta modulator (DM)



→ 1-bit (2-level) DPCM

→ Fixed Δ

→ slope $< \frac{\Delta}{T_s}$



message signal is compared with prev. quantised values and based on that go for $+\Delta$ or $-\Delta$.

↳ not done for every sample interval.

because Δ is fixed, look at slope and choose best Δ .

sinusoidal case $\Rightarrow m(t) = a_0 \sin(2\pi f_0 t)$

$$\frac{\Delta}{T_s} > 2\pi a_0 f_0$$

higher $\Delta \Rightarrow$ higher granular noise.

→ $\Delta = \text{diff b/w 2 Q.L.}$

$$\Delta > 2\pi a_0 f_0 T_s$$

stepsize should atleast be $2\pi a_0 f_0 T_s$ to avoid slope overload

on PCM

$$\sigma_n^2 = \frac{A^2}{12}$$

max^m amplitude $a_0 \leq \frac{\Delta}{2\pi f_0 T_s}$

based on this amplitude find (SNR)?

$$P_{\max} = \frac{a_0^2}{2} = \frac{\Delta^2}{8\pi^2 f_0^2 T_s^2}$$

(DM) quantisation noise power difference b/w 2 levels

$$\sigma_n^2 = \frac{(2\Delta)^2}{12} = \frac{\Delta^2}{3} = \frac{4\pi^2 a_0^2 f_0^2 T_s^2}{3}$$

= 2Δ (as you can go +Δ above and or -Δ below)

$$(SNR) = \frac{P_{\max}}{\sigma_n^2/3}$$

$$\frac{\Delta^2}{3} \Leftrightarrow [-f_s, f_s]$$

receiver $\Rightarrow$ LPF $\Rightarrow$ BW $\approx$ W

$+A \rightarrow 1$ 1 bit per

$-A \rightarrow 0$ sample signal power is in $-W$ to W .

encode

given m(t) to digital signal.

complete noise won't affect, only noise in $(-W, W)$ will affect

$$\frac{\frac{A^2}{3}}{2f_s} > 2W$$

$$P_{max} = \frac{3f_s^2}{8\pi^2 W f_0^2}$$

only way to change data rate $\rightarrow$ change sampling rate.

doubling sampling f_s $\rightarrow$ gives 9dB improvement

PCM $\Rightarrow$ 1 bit $\uparrow \Rightarrow$ 6dB SNR $\uparrow$

DM $\Rightarrow f_s \rightarrow 2f_s \Rightarrow$ 9dB SNR $\uparrow$
 $\uparrow 10 \log 8$

PCM
 40 kbps $\Rightarrow f_s = 8 \text{ kHz}$, 5 bits/sample
 80 kbps $\Rightarrow$ $\downarrow$ $\downarrow$ 10

each bit $\rightarrow$ 6dB improvement
 inc. 5 bits
 30dB improvement in SNR.

Delta modulator (no control over bits)

$\checkmark$ DM $8 \text{ kHz} \times 1 \text{ bit/sample} = 8 \text{ kbps}$

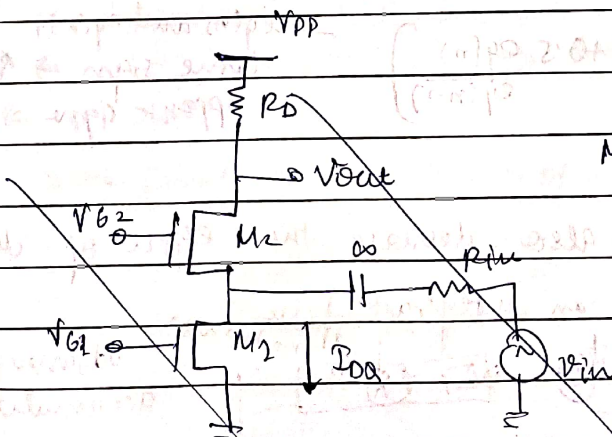
inc. sampling rate $\rightarrow 8 \text{ kHz} \times 2 = 16 \text{ kbps}$
 to improve

improvement in PCM is $>$ DM

Limitation of DM $\Rightarrow$ not good for slowly varying part
 good for steep slope areas.

$\Delta \rightarrow \Delta(n)$ adaptive delta modulator.

08/02/2026



both transistors have to be in saturation.

$M1 \rightarrow$ in saturation $\rightarrow$ good source

$M2 \rightarrow$ " $\rightarrow$ to work in common source amplifier.

$$P_{Da} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{G1} - V_{TH})^2$$

$$V_{G1} = V_{G2}$$

Now to choose V_{G2} ?

Adaptive DM

$$\Delta(n) = |A(n-1)| e(n) + A(n) e(n-1)$$

09/02/2026

Delta modulator:

problem: input part $\rightarrow$ input - output of the delay

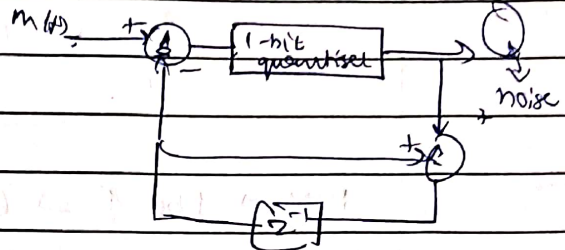
difference in discrete domain

is approximation to derivative and of message signal

noise will have high impact due to the summation.

quantiser $\Rightarrow$ thresholding.

delay inverse $\Rightarrow$ delay.



we introduce a step size logic control.

Adaptive DM:

$$A \geq \max \left| \frac{dm(t)}{dt} \right|$$



1) discrete type

$\rightarrow$ set of 4 values stored in lookup table

2) continuous type

If Δ is decided looking at high slope region $\Rightarrow$ not good for the small slope region and if decided by looking at low slope region $\Rightarrow$ not good for high slope region. So the Δ should be adaptive.

1)

a) constant type $\Rightarrow$ values are stored in lookup table

b) varying type $\Rightarrow$ values are stored in lookup table but they keep varying.

$$\Delta(n) = g(n) \Delta(n-1)$$

where $g(n) = \begin{cases} k & \text{if } e_q(n) = e_q(n-1) \\ 1/k & \text{if } e_q(n) \neq e_q(n-1) \end{cases}$



if it keeps on inc, $+A, +A, +A$ but if it keeps on changing like $+A, -A$.

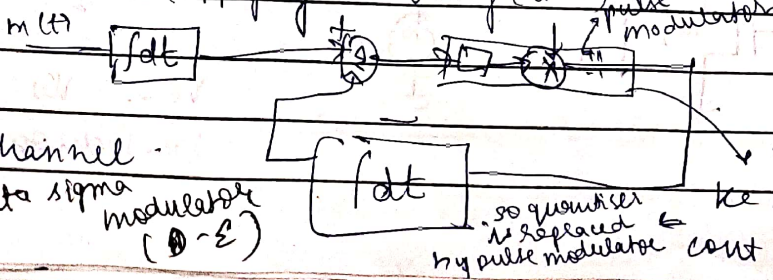
cont type.

$$\Delta(n) = \begin{cases} \Delta(n-1) + \left[1 + 0.5 \frac{e_q(n)}{e_q(n-1)} \right] \Delta_{min} & \text{if } e_q(n) \text{ and } e_q(n-1) \text{ have same sign} \\ \Delta(n-1) - \left[1 + 0.5 \frac{e_q(n)}{e_q(n-1)} \right] \Delta_{min} & \text{if } e_q(n) \text{ and } e_q(n-1) \text{ have opposite sign} \end{cases}$$

$\rightarrow e_q(n)$ and $e_q(n-1)$ have same sign $\Rightarrow \uparrow$ by 50%
opposite sign $\Rightarrow \downarrow$ by 50%

we want to use DM but also decrease the effect of difference operator. (Applying an integrator)

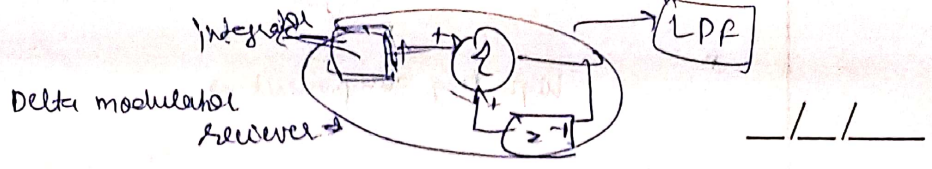
problem is impact of noise in channel.



Delta sigma modulator (0-1)

so quantiser is replaced by pulse modulator

advantage of accumulation operator
high sampling rate
keeps the signal goes into const. time scenario



integration in time domain $\Rightarrow$ low freq. components are emphasized
 now block gets multiplied
 due to integrator operator.

best case scenario but adding at receiver's side.

$$\{ (x(t) - x(t-1)) + n(t) \} + x'(t-1) \Rightarrow$$

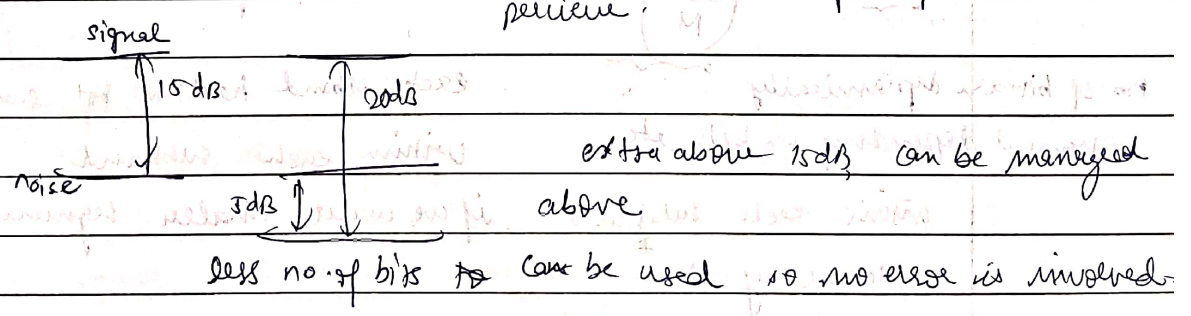
 improving this over this
 so that don't get highlighted much.

$$\int x(t) dt - \int () dt$$

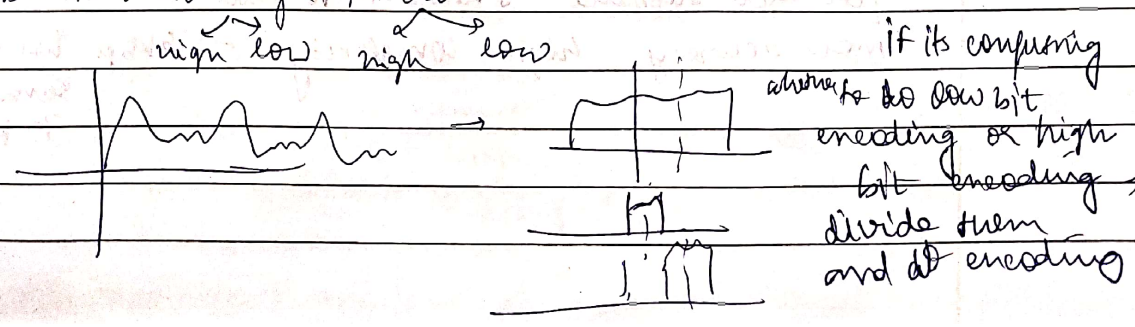
11/02/2026 low bitrate encoding of speech and video signals.

- 1) Adaptive PCM (time domain) $\rightarrow$ ADPCM $\rightarrow$ 32 kbps
 - 2) Adaptive sub-band coding (frequency domain) (ASBC)
 - a) Periodicity in voiced speech
 - b) Noise masking.
- on human speech:
 (A) interrupted $\rightarrow$ 16 kbps
 (S) not interrupted $\rightarrow$ 32 kbps
 1) Voiced segment
 2) unvoiced segment
 3) silence segment

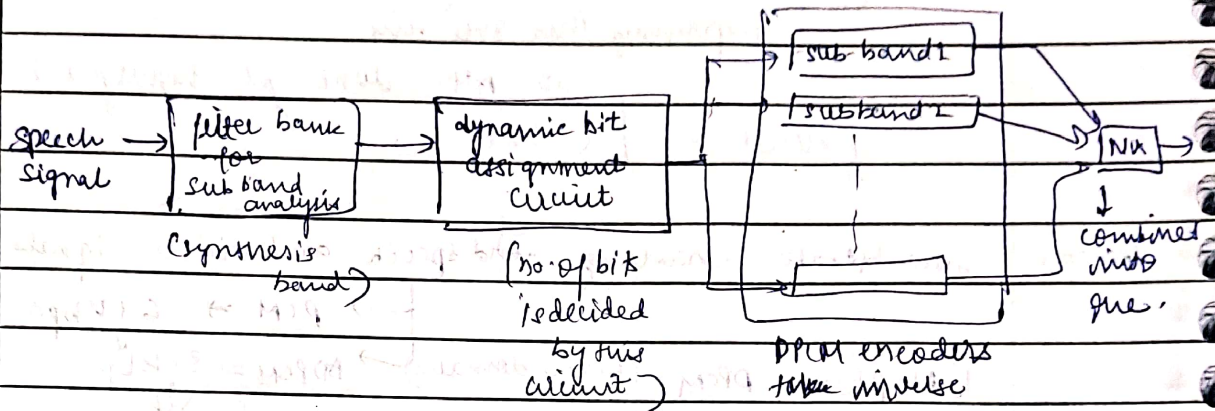
diff b/w power level and noise $> 15\text{dB}$
 $\Rightarrow$ human ear can't perceive.
 being exploited.



Signal power low $\Rightarrow$ more no. of bits. high $\Rightarrow$ less bits.
 break into high & low



high freq components $\Rightarrow$



exactly opposite in receiver side

demux $\rightarrow$ PCM decoder

PT do encoding of each sub band.

PCM encoding.

eg $f_s = 8\text{kHz}$

$N = 2\text{ bits/sample}$

analysis band in receiver side instead of synthesis band.

$M = 4$

$f_s = 8\text{kHz}$

$N = 2\text{ bits/sample}$

$$\text{Bitrate} = N f_s \Rightarrow \left(\frac{f_s}{M} \right) (MN)$$

no. of stages I have $\Rightarrow$ no. of bands

no. of bits are dynamically

changed depending on bits, etc.

each subband has (MN) bit sample

within each subband

within each subband, if we want smaller segments,

accuracy $\Rightarrow$

low amplitude $\Rightarrow$ more no. of bits

high amplitude $\Rightarrow$ less no. of bits

for each subband I have MN bits.

more accuracy, high complexity $\Rightarrow$ 16 kbps has quality same as 32 kbps.

Mos $\Rightarrow$ qualitative score

1/1

If SNR deteriorates, it does not affect the perception quality

Mos $\rightarrow 0 \rightarrow 5$ mean order score
 PCM 64 kbps
 ADPCM 32 kbps
 ATBC 16 kbps
 $\rightarrow$ Mos ≈ 4

Sampling and quantisation $\rightarrow$ Nyquist (CS) $\rightarrow$ DPCM, DTM

Bandpass sampling $\rightarrow$ (skelai / proakis)

Derivation of Nonuniform quantisation $\rightarrow$ Nyquist (DC)

Multiplexing (Carson)

16/02/26

Adaptive sub-band coding:

Overall rate has to be same !!!

We are just trying better encoding

$f_s = 8 \text{ kHz}$
 $M = 4 \rightarrow$ no. of subbands

$N = 2$

each subband needs to be coded.

(equal importance to all subbands)

does not look at the content

within the subband look at the power component

Noise masking ?

$N_{fs} = 2 \times 8 = 16 \text{ kHz}$

encoding is diff. because

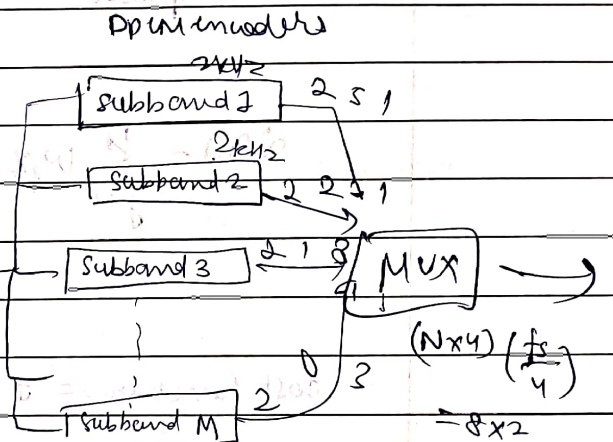
we are assigning diff. no. of bits to high & low freq.

high freq. components are of higher significance

assign more bits to them

low freq component are of higher signif

assign more bits



Adaptive

looks at content of subband

Conventional sub band coding
 each band is given 2 kHz

does not look

at the content

high freq. components are of higher significance

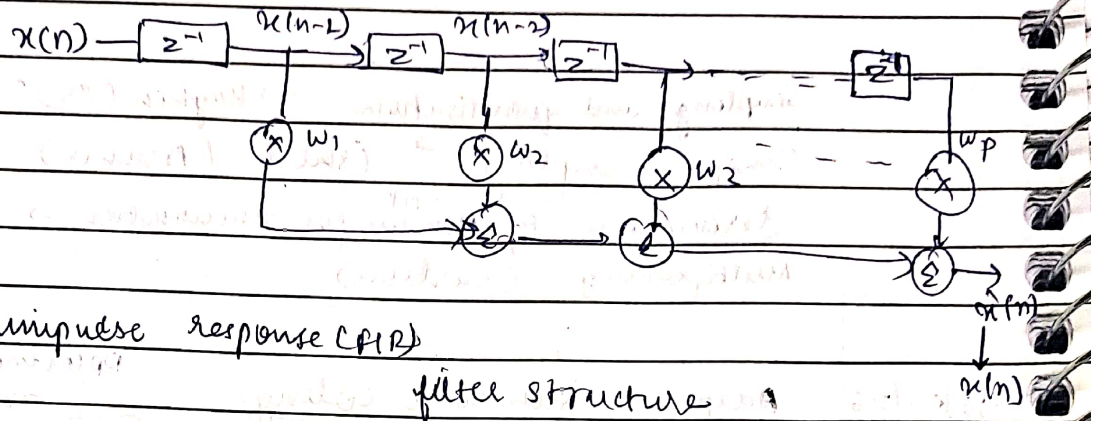
assign more bits to them

low freq component are of higher signif

assign more bits

Adaptive prediction: (ADPCM)

Linear prediction:



Finite impulse response (FIR)

filter structure

$$\hat{x}(n) = \sum_{j=1}^p w_j x(n-j)$$

$\hat{x}(n)$ should approach $x(n)$

$$J = (x(n) - \hat{x}(n))^2$$

that will depend on $w_1, w_2, w_3, \dots$

$$\text{cost function} = J = E[e^2(n)]$$

$$= E[(x(n) - \hat{x}(n))^2]$$

$$= E[x(n)^2] + E[\hat{x}(n)^2] - 2E[x(n)\hat{x}(n)]$$

$$= E[x(n)^2] + E\left[\sum_{j=1}^p w_j x(n-j) \sum_{k=1}^p w_k x(n-k)\right]$$

$$- 2E\left[x(n) \sum_{j=1}^p w_j x(n-j)\right]$$

$$\frac{\partial J}{\partial w_i} = 0 + 2E\left[\sum_{j=1}^p w_j x(n-j) x(n-i)\right] - 2E[x(n)x(n-i)] = 0$$

$$\Rightarrow E\left[\sum_{j=1}^p w_j x(n-j) x(n-i)\right] = E[x(n)x(n-i)]$$

$$\vec{w} = [w_1 \ w_2 \ \dots \ w_p]^T$$

$$R_x(p)$$

$$\vec{r} = [R_x(1) \ R_x(2) \ \dots \ R_x(p)]^T$$

$$\vec{R}_x = \begin{bmatrix} R_x[0] & R_x[1] & \dots & R_x[p-1] \\ R_x[1] & & & \\ \vdots & & & \\ R_x[p-1] & & & R_x[0] \end{bmatrix}$$

$$\vec{R}_x \vec{w}_0 = \vec{r}_x \Rightarrow \vec{w}_0 = \vec{R}_x^{-1} \vec{r}_x$$

expectation

take lot

of experiments

& taking mean

wiener - hopff solution

$$J_{min} = \sigma_x^2 - \vec{r}_x^T \vec{R}_x^{-1} \vec{r}_x$$

$$J_{min} < \sigma_x^2$$

always true

inverse operation
req. lot of

calculation

we will deal with 2 problems when we move from
linear predⁿ to adaptive predⁿ

18/02/26

$$\vec{w}_0 = \vec{R}_x^{-1} \vec{r}_x$$

wiener hay solution

→ E[]

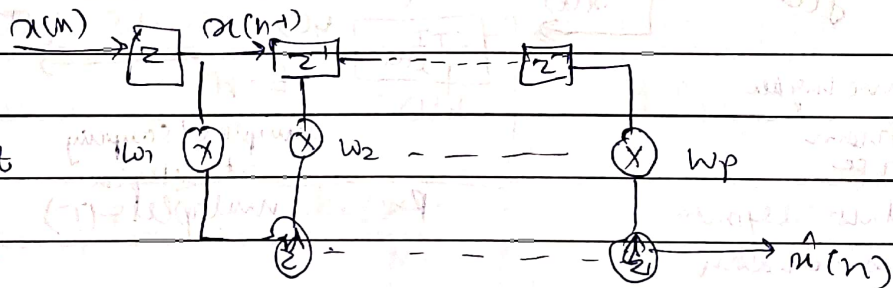
Method of steepest descent

→ E[]

→ Matrix

consider 2 taps

we go against the gradient but want to go as fast as we can
1st order filter
to optim soln



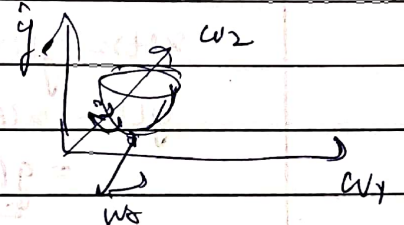
say when you start
your process

from w1 & w2,
first error eg.
i get while

one the curve

$$J = E[e^2(n)]$$

$$J = e^2(n) = [x(n) - \hat{n}(n)]^2$$



initially we

want bigger

jumps

then later
smaller
jumps.

$$w_k(n+1) = w_k(n) - \frac{1}{2} x(n) e(n) \nabla J(n)$$

given value of w, w,
find gradient at
every point & go
against the gradient
(initially large steps then
small steps)

||||

expectation operator is approximated as instantaneous operator.

—|—|—

$$\nabla^2 f(n) = \frac{\partial^2 f(n)}{\partial w_k} = 2 \left[\sum_j w_j x(n-j) x'(n-k) \right] - 2 \left[x(n) x'(n-k) \right]$$

$$e^2(n)$$

$$w_k(n+1) = w_k(n) - \frac{\mu(n)}{2} \left[\sum_j w_j x(n-j) x(n-k) - x(n) x(n-k) \right]$$

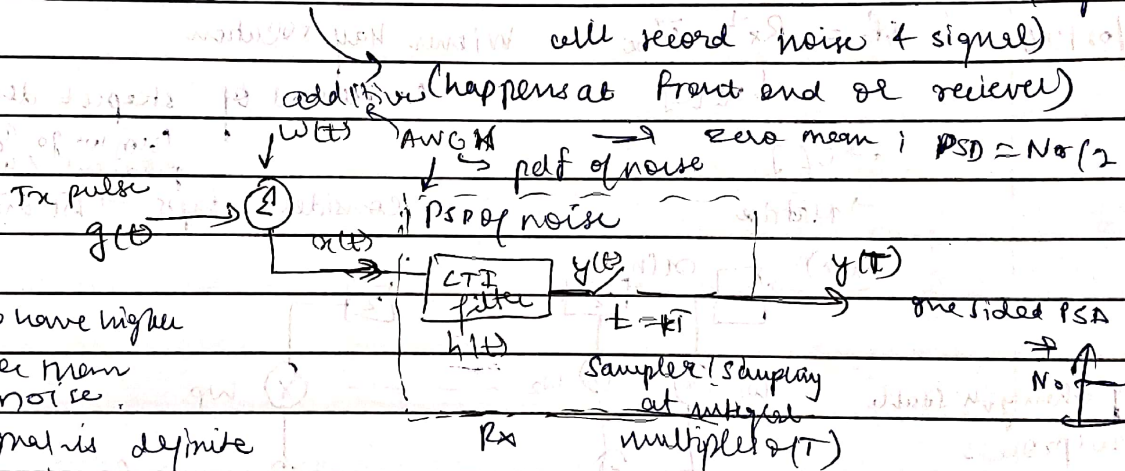
$$= w_k(n) - \mu(n) x(n-k) \left[-\sum_j w_j x(n-j) \right]$$

whatever estimate is received $\Rightarrow$ put in DPCM

Baseband data transmission:

Sources of bit error:

- 1) Inter-symbol interference
- 2) channel noise (received at receiver, front end of signal)



Signal to have higher power than noise.

But signal is definite & noise is random

$$x(t) = g(t) + w(t)$$

$$y(t) = x(t) * h(t)$$

$$= \underbrace{g(t) * h(t)}_{g_o(t)} + \underbrace{w(t) * h(t)}_{n(t)}$$

$$SN(f) = \frac{N_0 |H(f)|^2}{2}$$

LTI response we need to maximize signal to noise ratio

should be similar to $g(t)$

Matched filter $\propto g(t)$

i know what kind of waveform is used e.g. gaussian etc value at any point is not known

$$\eta = \frac{|g_o(t)|^2}{E[n^2(t)]}$$

$$\begin{aligned} g_o(t) &= g(f) * h(t) \\ &= \text{IFT} \left\{ G(f) H(f) \right\} \\ &= \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f t} df \end{aligned}$$

$$|g_o(t)|^2 = \left| \int_{-\infty}^{\infty} G(f) H(f) e^{j2\pi f t} df \right|^2$$

$$[n^2(t)] = \int_{-\infty}^{\infty} S_n(f) df = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df$$

19/02/26

$$\eta = \frac{\left| \int_{-\infty}^{\infty} H(f) G(f) e^{j2\pi f t} df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Schwarz inequality:

$$\phi_1(x), \phi_2(x)$$

$$\int_{-\infty}^{\infty} \phi_1^2(x) dx < \infty \quad \int_{-\infty}^{\infty} \phi_2^2(x) dx < \infty$$

$$\left| \int \phi_1(x) \phi_2(x) dx \right|^2 \leq \int |\phi_1(x)|^2 dx \int |\phi_2(x)|^2 dx$$

equality holds when if and only if $\phi_1(x) = k \phi_2(x)$

$$\text{let } \phi_1(x) = H(f) \quad \phi_2(x) = G(f) e^{j2\pi f t}$$

$$\eta = \frac{\left| \int_{-\infty}^{\infty} H(f) G(f) e^{j2\pi f t} df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$\eta \leq \frac{2}{N_0} \int_{-\infty}^{\infty} |G(f) e^{j2\pi f t}|^2 df$$

$$\eta \leq \frac{2}{N_0} \int_{-\infty}^{\infty} |G(f)|^2 df$$

$$\boxed{\eta \leq \frac{2E}{N_0}} \Rightarrow \eta_{\max} = \frac{2E}{N_0} \text{ if } H(f) = k G^*(f) e^{-j2\pi f t} \text{ opt.}$$

$$h_{opt}(t) = \int_{-\infty}^{\infty} k G^*(f) e^{-j2\pi fT} e^{j2\pi f t} df$$

$$= k \int_{-\infty}^{\infty} G^*(f) e^{-j2\pi f(t-T)} df$$

for real $g(t)$ $G^*(f) = G(-f)$

$$h_{opt}(t) = k \int G(-f) e^{-j2\pi f(t-T)} df$$

$$h_{opt}(t) = k g(T-t) \rightarrow \text{time reversed and shifted version of } g(t)$$

shape of matched signal does not impact max. SNR.

correlator $\Leftarrow$ filter has been matched to original signal $g(t)$

is important

not the shape

(go through properties of matched filter)

why is it true

A matched filter can be replaced by a correlator & T/F?



at the receiver side correlator is representing a matched filter.

Bit error rate due to channel noise:

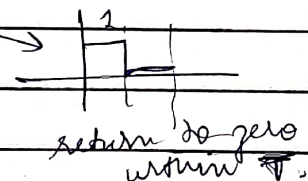
Polar $\rightarrow$ $1 \rightarrow A$, $0 \rightarrow -A$ unipolar $\rightarrow$ $1 \rightarrow A$, $0 \rightarrow 0$



NRZ

non return to zero

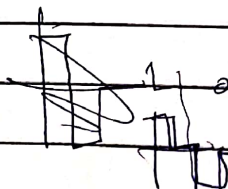
return to zero



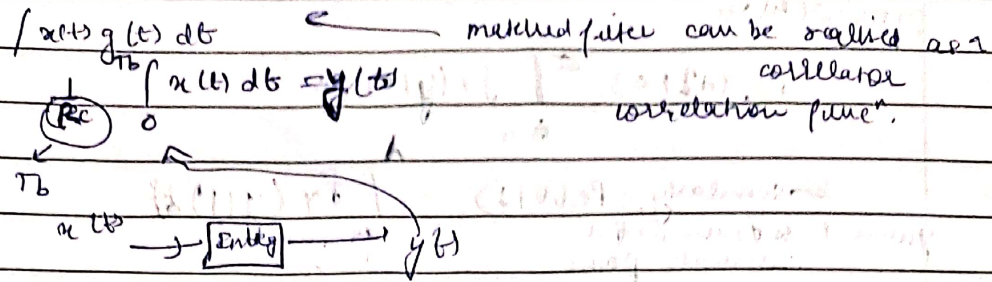
Polar NRZ

$1 \rightarrow A$
 $0 \rightarrow -A$

$$x(t) = \begin{cases} A + w(t), & 1 \\ -A + w(t), & 0 \end{cases}$$



input is $x(t)$, correlating with $x(t)$
 $\rightarrow (x(t))^2$



mean $\Rightarrow -A$

variance?

$$y(t) = \frac{1}{T_b} \int_0^{\tau_b} x(t)dt$$

Bit 0 is transmitted $\Rightarrow x(t) = -A + w(t)$

$$y(t) = -A + \frac{1}{T_b} \int_0^{\tau_b} w(t)dt$$

the r.v $y \rightarrow$ pdf?

y will be a gaussian pdf with mean $(-A)$, variance = ?

$$\sigma_y^2 = E[(y - (-A))^2]$$

$$= E \left[\left(\frac{1}{T_b} \int_0^{\tau_b} w(t)dt \right)^2 \right]$$

auto correlation delay by t and u

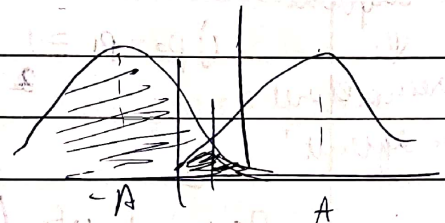
$$= \frac{1}{T_b^2} \int_0^{\tau_b} \int_0^{\tau_b} E[w(t)w(u)] dt du$$

$$R_w(t, u) = \frac{N_0}{2} \delta(t - u)$$

$$= \frac{N_0}{2T_b^2} \int_0^{\tau_b} \int_0^{\tau_b} \delta(t - u) dt du = \frac{N_0}{2T_b}$$

conditional pdf
 given bit 0 is transmitted what is the conditional pdf of y .

$$f_y(y|0) = \frac{1}{\sqrt{N_0/2}} e^{-\frac{(y+A)^2}{N_0/2}}$$



when error $\Rightarrow$ received y

$\tau =$ threshold

(threshold as symmetry)

comes in overlapping region beyond τ , category is clear.

conditional prob that 0 is transmitted & is received.

prob. of error

$$P_e(1|0) = \int_{A_1}^{\infty} f_Y(y|0) dy$$

similarly $P_e(0|1) = \int_{-A_1}^{-\infty} f_Y(y|1) dy$

given 1 is transmitted
calculate prob =

20/02/26

$$f_Y(y|0) = \frac{1}{\sqrt{\pi N_0/T_b}} e^{-\frac{(y+A)^2}{N_0/T_b}}$$

$$P_{10} = \int_{A_1}^{\infty} f_Y(y|0) dy = \frac{1}{\sqrt{\pi N_0/T_b}} \int_{A_1}^{\infty} e^{-\frac{(y+A)^2}{N_0/T_b}} dy$$

$$P_{01} = \frac{1}{2} \operatorname{erfc}\left(\frac{A-A_1}{\sqrt{N_0/T_b}}\right) = \frac{1}{2} \operatorname{erfc}\left(\frac{A+A_1}{\sqrt{N_0/T_b}}\right)$$

$P_0 \rightarrow$ A priori probability of sending '0'.

$P_1 \rightarrow$ A priori probability of sending '1'.

Complementary error function $\Rightarrow \operatorname{erfc}(u) = \frac{2}{\sqrt{\pi}} \int_u^{\infty} e^{-z^2} dz$
 $\frac{d}{du} \operatorname{erfc}(u) = -\frac{2}{\sqrt{\pi}} e^{-u^2}$

Avg probability of error = $P_0 P_{10} + P_1 P_{01}$

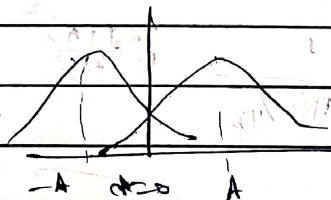
$$P_e = \frac{P_0}{2} \operatorname{erfc}\left(\frac{A+A_1}{\sqrt{N_0/T_b}}\right) + \frac{P_1}{2} \operatorname{erfc}\left(\frac{A-A_1}{\sqrt{N_0/T_b}}\right)$$

If a priori probabilities are different

take partial derivative wrt A , find out the derivative value of A .

threshold will be different

1) $P_0 = P_1 = \frac{1}{2}$



$$A_{opt} = \frac{N_0}{4A T_b} \ln\left(\frac{P_0}{P_1}\right)$$

$$P_e = \frac{1}{2} \operatorname{erfc}\left(\frac{A}{\sqrt{N_0/T_b}}\right)$$

$A^2 T_b \Rightarrow$ energy of pulse.

only for

white noise,

$$= \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{A^2 T_b}{N_0}}\right)$$

$$P_e \Leftrightarrow E_b/N_0$$

there are cases

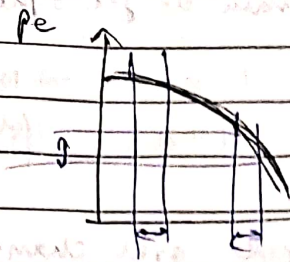
when noise

is not white

$$= \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

(depends on energy not the shape)

if $E_b/N_0 \uparrow \Rightarrow P_e \downarrow$



ESD bit energy to noise ratio
 $\left(\frac{E_b}{N_0}\right)$ in dB
 discrete version of (SNR)

increasing $\frac{E_b}{N_0}$ at lower power

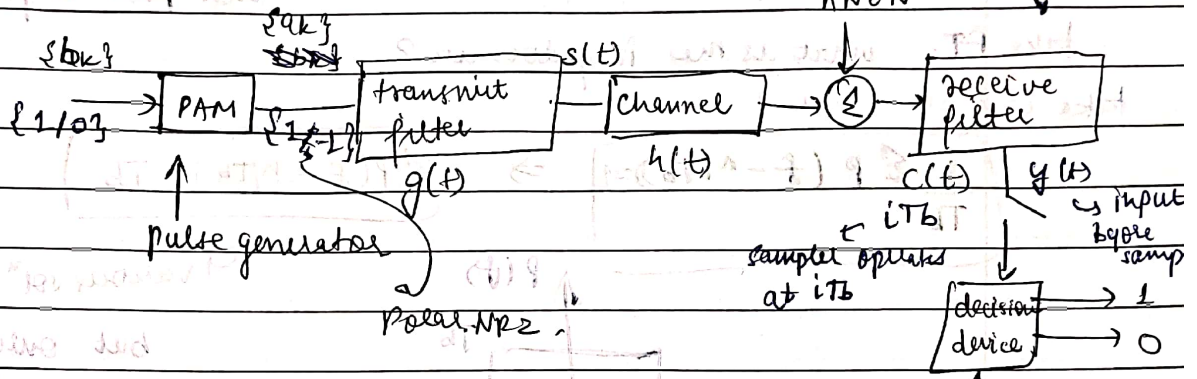
is easy (hardware)

but at higher power is difficult

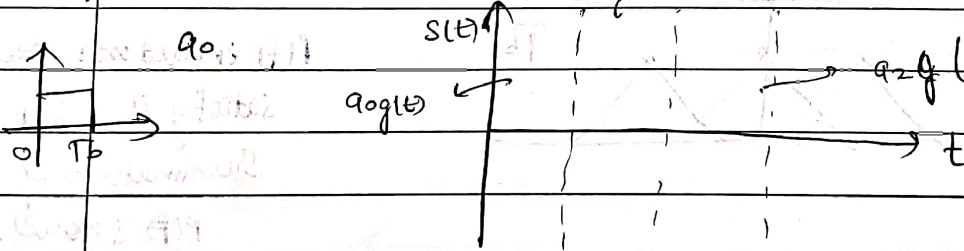
(at higher power it is more beneficial)

Inter symbol interference

conventionally we use polar NRZ



output of receive filter we are going to neglect the channel noise part. and take signal part only!



$$s(t) = \sum a_k g(t - kT_b)$$

$$p(t) = g(t) * h(t) * c(t)$$

$$y(t) = \sum a_k p(t - kT_b) + n(t)$$

$$t_i = iT_b$$

$$n(t) = w(t) * c(t)$$

$$y(t_i) = \sum_k a_k (iT_b - kT_b) + n(t_i) \quad \text{this has to be 0}$$

$$= a_i p(0) + \sum_{k, k \neq i} a_k p(iT_b - kT_b) + n(t_i)$$

desired signal component

ISI

intersymbol interference

$\hookrightarrow p(0)$ has to be 1

$$a_k p(iT_b - kT_b) = 0 \quad \Rightarrow \quad p(iT_b - kT_b) = 0$$

impulse sampling at iT_b
 this is samples in t -domain, in f -domain we get replicas.

$$p((i-k)T_b) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases} \rightarrow \text{Nyquist criterion for zero ISI}$$

time domain condⁿ.

but I don't have any control over channel.

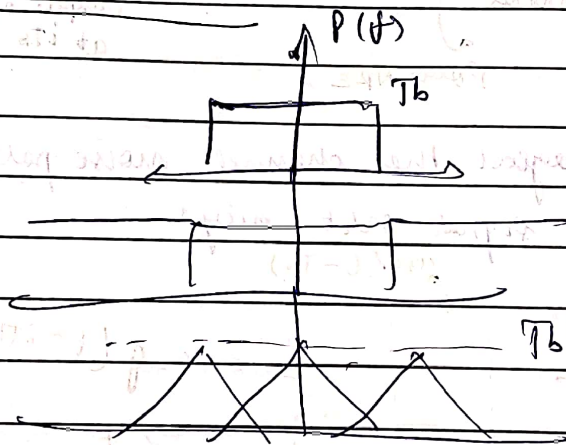
transmit & receive filter are exact replicas of each other.

so we know the channel, the complete T.F, we can find the transmit & receive filter and put it for both.

take P.T. what is the freq. domain?

take $i=k$ and $i \neq k$

$$\frac{1}{T_b} \sum P(f - n/T_b) = 1 \Rightarrow \sum P(f - n/T_b) = T_b$$



various solⁿ for $P(f)$, but only condition req^d is that summation of all replicas is T_b .

$P(f)$ should not necessarily satisfy T_b , but summation of $P(f)$ should be T_b .

how we are going to design transmit and receive filter st. the T_b condⁿ is met.